

8. [Maximum mark: 6]

A continuous random variable X has a probability density function f given by

$$f(x) = \begin{cases} kx & 0 \leq x \leq k \\ 2kx - x^2 & k < x \leq 2k \\ 0, & \text{otherwise} \end{cases}$$

where $k > 0$.

(a) Show that k satisfies the equation $7k^3 = 6$.

[2]

(b) Find the median of X .

[4]

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